

SURIM OH

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RESEARCH INTEREST

My research lies in **computer architecture, systems, and datacenter workloads**, working both sides of the architecture-workload loop: characterizing emerging workload classes at the CPU cycle level, and designing microarchitectural mechanisms that target the bottleneck.

1. Run-ahead control in the decoupled frontend / FDIP path. UDP (Utility-Driven FDIP) learns the performance impact of run-ahead distance and the usefulness of instruction-cache lines, then drives optimal run-ahead and i-cache filtering.

2. Post-recovery instruction supply. JUMP (Junction-based Unified Miss-Point Prefetching) trains branch→i-cache-miss-point pairs online from branch prediction recoveries and launches a secondary FDIP thread directly at the downstream i-cache miss-point, reusing existing branch-predictor structures.

3. Agentic AI as a new architectural target. The first cycle-level CPU characterization of agentic AI workloads, across seven Python agent frameworks plus a deployed production CLI agent, and a heterogeneous-core architectural response motivated by the resulting per-phase profile.

I am also broadly interested in **CPU performance modeling and simulation methodology**: microarchitecture simulators, sampling techniques, deterministic replay, and PMU-driven characterization of multi-core / multi-thread / heterogeneous CPU-GPU workloads. During industry internships I worked on **ASIC architecture for LLMs**, including chiplet-based accelerators and wafer-scale design.

EDUCATION

University of California, Santa Cruz

PhD in Computer Science and Engineering

Santa Cruz, CA, USA

Sep 2020 - Jun 2026

· Advisor: Professor Heiner Litz, Total GPA: 4.0/4.0

· Thesis: *Efficient Instruction Supply for Datacenter Processors* (defense: May 2026)

Seoul National University

Master of Science in Computer Science and Engineering

Seoul, South Korea

Feb 2015 - Feb 2017

· Advisor: Professor Bernhard Egger, Total GPA: 3.52/4.0

· Thesis: *Hierarchical Manycore Resource Management Framework using Control Processors* [pdf]

Sogang University

Bachelor of Science in Computer Science and Engineering

Seoul, South Korea

Feb 2011 - Feb 2015

· Total GPA: 3.70/4.0 (95.8/100), Summa Cum Laude

· Exchange student at **Northern Arizona University**

AZ, USA, Spring 2014

WORK EXPERIENCE

Meta

ASIC Engineer Intern, Architecture

Sunnyvale, California, USA

June 2025 - Sep 2025

Meta

ASIC Engineer Intern, Architecture

Sunnyvale, California, USA

June 2024 - Sep 2024

Akeana

PhD Intern

San Jose, California, USA

June 2023 - Sep 2023

SAP Labs Korea
Developer
Hyundai Motor Company R&D Division
Engineer

Seoul, South Korea
Jan 2018 - Sep 2020
Hwaseong, South Korea
Feb 2017 - Jan 2018

SKILLS

C, C++, Python, Bash, Scarab simulator, DynamoRIO, SimPoint, Docker, Slurm, Linux `perf`, McPAT, Intel PCM, LLVM, Verilog, Magic, OpenCL, HSIM-TQSIM (SystemC-based) manycore simulator

PUBLICATIONS

MICRO 2025: Yinyuan Zhao, Surim Oh, Mingsheng Xu, Heiner Litz. ATR: Out-of-Order Register Release Exploiting Atomic Regions. *In the 58th Annual IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 2025. [pdf]

HSCC 2025: Paul K. Wintz, Yasin Sonmez, Paul Griffioen, Mingsheng Xu, Surim Oh, Heiner Litz, Ricardo G. Sanfelice, Murat Arcak. SHARC: Simulator for Hardware Architecture and Real-time Control. *In Proceedings of the 28th ACM International Conference on Hybrid Systems: Computation and Control (HSCC)*, 2025. [pdf]

HotInfra 2024: Surim Oh, Eric Qin, Yang Yang, Mengchi Zhang, Raj Parihar, Ashish Pandya. LLM Inference Performance on Chiplet-based Architectures and Systems. *for Poster Presentation In the 2nd Workshop on Hot Topics in System Infrastructure (HotInfra) Co-located with SOSP*, 2024. [pdf]

ISCA 2024: Surim Oh, Mingsheng Xu, Tanvir Ahmed Khan, Baris Kasikci, Heiner Litz. UDP: Utility-Driven Fetch Directed Instruction Prefetching. *In the 51st International Symposium on Computer Architecture (ISCA)*, 2024. [pdf]

M.S. Thesis: Surim Oh. Hierarchical Manycore Resource Management Framework using Control Processors. *Seoul National University*, Seoul, South Korea, February 2017. [pdf] [slides]

TPDS 2019: Younghyun Cho, Surim Oh, and Bernhard Egger. Performance Modeling of Parallel Loops on Multi-Socket Platforms using Queueing Systems. *In IEEE Transactions on Parallel and Distributed Systems (TPDS)*, in press, available online, September 2019. [pdf]

MULTIPROG 2017: Younghyun Cho, Surim Oh, and Bernhard Egger. Cooperative Parallel Runtimes for Multicores. *Presented at the 10th International Workshop on Programmability and Architectures for Heterogeneous Multicores*, January 2017. [pdf]

CATC 2016: Surim Oh, Younghyun Cho, and Bernhard Egger. Efficient Resource Management for Many-cores with Centralized L2 Caches using Distributed Control Processors. *Presented at the 7th Compiler, Architectures and Tools Conference*, September 2016. [pdf] [slides]

PACT 2016: Younghyun Cho, Surim Oh, and Bernhard Egger. Online Scalability Characterization of Data-parallel Programs on Many Cores. *In Proceedings of the 25th International Conference on Parallel Architectures and Compilation Techniques (PACT)*, September 2016. [pdf]

JSSPP 2016: Younghyun Cho, Surim Oh, and Bernhard Egger. Adaptive Space-shared Scheduling for Shared-memory Parallel Programs. *Presented at the 20th Workshop on Job Scheduling Strategies for Parallel Processing*, May 2016. *In Lecture Notes in Computer Science (LNCS)*, Volume 10353, July 2016. [pdf]

PATENTS

US Patent 2018A: Bernhard Egger, Surim Oh, Younghyun Cho, Dong-hoon Yoo. Method of processing OpenCL Kernel and Computing Device Therefor. *US Patent 20180181443A1*, June 2018. Worldwide applications in KR, EP, CN, JP including US.

US Patent 2018B: Bernhard Egger, Younghyun Cho, Surim Oh, Dong-hoon Yoo. Computing devices and methods of allocating power to plurality of cores in each computing device. *US Patent 20180246554A1*, August 2018. Worldwide applications in KR, CN including US.

RESEARCH AND PROJECT EXPERIENCE

University of California, Santa Cruz

PhD Student

Santa Cruz, CA, USA

Sep 2020 - Present

- Studied a state-of-the-art **Fetch Directed Instruction Prefetching (FDIP)** on a **CPU microprocessor** simulator, Scarab, and the performance impact of FDIP on frontend-bound applications with large instruction footprints.
- Introduced **Utility-Driven FDIP (UDP)** by learning the performance impact of running-ahead distance of FDIP and the usefulness of instruction cache lines for optimal running-ahead distance and filtering mechanisms - *ISCA 2024*.
- Designed **JUMP** (Junction-based Unified Miss-Point Prefetching), a follow-on FDIP mechanism that trains trigger→miss-point pairs online from prediction-recovery events (branch mispredictions and BTB misses) using a small entanglement table, then launches a secondary FDIP thread directly at the downstream instruction-cache miss-point whenever a trained trigger is predicted, reusing existing branch-predictor tables. Identified five recurring prediction-failure scenarios that together cover most i-cache miss cycles in datacenter and Google workloads - *in submission*.
- Building the **first cycle-level microarchitectural characterization of agentic AI CPU workloads** across seven Python agent frameworks (LangChain, LangGraph, AutoGen, Haystack, SWE-Agent, Toolformer, ChemCrow). Instrumented each framework with phase-boundary markers and PC-sampled call-stack attribution to identify shared orchestration overhead patterns. Further proposing a **heterogeneous-core architectural response** for the agent CPU workload class, evaluated end-to-end on the Scarab simulator with extensions for multi-region ROI-aware simulation and per-region energy attribution - *in preparation*.
- Contributed substantial extensions to the **Scarab cycle-accurate simulator** [github] (decoupled frontend + FDIP, instruction prefetchers, multi-level branch predictor, multi-level BTB, ROI marker handling).
- Founded and led **Scarab-infra** [github], the open-source workflow tool used by Litz Lab (and external researchers) to orchestrate Scarab cycle-accurate simulations end-to-end through a unified command-line interface. Designed the descriptor-based experiment model that drives Scarab build, simulation, trace collection, statistics aggregation, and visualization, and the Docker + Slurm integration for parallel SimPoint and ROI workflows. Authored the initial commit and continued as sustained maintainer, adding end-to-end CI, Dockerized perf, DynamoRIO tracing, parallel simulations, memory tracking, and descriptor schema validation.

Meta

ASIC Engineer Intern, Architecture

Sunnyvale, CA, USA

Jun 2025 - Sep 2025

- Wafer-scale architecture modeling and exploration for future MTIAs.
- Developed a flexible and highly configurable transaction-based simulator to model wafer-scale architectures with different parallelism strategies (TP/PP) for LLM inference. Performance analysis of different feasible wafer-scale architectures.

Meta <i>ASIC Engineer Intern, Architecture</i>	<i>Sunnyvale, CA, USA</i> <i>Jun 2024 - Sep 2024</i>
- Performance projection of LLM Inference on NVIDIA H100, B200, and two different versions of MTIAs by exploiting an open-source hardware evaluation framework, LLMCompass. - Detailed analysis on chiplet-based architecture has been accepted as a poster to a workshop HotInfra'24 co-located with SOSP 2024.	
Akeana <i>PhD Intern</i>	<i>San Jose, CA, USA</i> <i>Jun 2023 - Sep 2023</i>
- Designed and implemented a trace cache storing consecutively executed basic blocks on an internal RISC-V ISA simulator on top of Spike. - The designed trace cache with branch history and confidence counter obtains performance gain by expanding fetch bandwidth.	
SAP Labs Korea <i>Developer</i>	<i>Seoul, South Korea</i> <i>Jan 2018 - Sep 2020</i>
- Contributed to table replication technology of SAP HANA DBMS to scale out mixed OLTP/OLAP workloads. - Designed and implemented replication log formats and protocols for querying the logs from other DB systems to be compatible with other general DB systems with minimum source side cost.	
Hyundai Motor Company R&D Division <i>Engineer</i>	<i>Hwaseong, South Korea</i> <i>Feb 2017 - Jan 2018</i>
- Contributed to Vehicle data monitoring system that collects in-vehicle data from many distributed ECUs and store the data in a remote centralized server. - Designed and implemented a merge tool for video data obtained from testing cars to develop Advanced Driver Assistance Systems technology.	
Computer System And Platform Laboratory , Seoul National University <i>Graduate Researcher</i>	<i>Seoul, South Korea</i> <i>Feb 2015 - Feb 2017</i>
- Worked on resource management techniques on manycore SoC architecture for simultaneously running OpenCL applications, distributing/minimizing runtime scheduling overhead and power consumption. The technique exploits architecture support of tiny/hierarchical control processors on top of a TQSIM-HSIM (Timed QEMU-based & SystemC-based) manycore simulator. - CATC 2016, M.S. Thesis, US Patent 2018A, US Patent 2018B. - Collaborated with four other research groups from SNU and Samsung Advanced Institute of Technology and presented at HumanTech Paper Award organized by Samsung (not awarded). - Developed a tool that queries HW performance counters related to the NUMA interconnection network in Intel/AMD systems for an analytical performance model on NUMA architecture based on queueing theory to estimate the resource utilization on multiprocessor systems for parallel programs - PACT 2016, TPDS 2019.	
AWARD	
CSE Excellent PhD Award 2026 University of California, Santa Cruz	<i>June 2026</i>
MICRO'25 Student Travel Grant 58th IEEE/ACM International Symposium on Microarchitecture	<i>November 2025</i>
SnuMAP: 2nd prize in 10th Open Source Software World Challenge SNU Manycore Profiler for Big-Data with SNU team	<i>Seoul, South Korea</i> <i>December 2016</i>

TEACHING EXPERIENCE

Advanced Computer Architecture (CSE220), UC Santa Cruz *Fall 2024*
Teaching Assistant

Computer Architecture (CSE120), UC Santa Cruz *Fall 2023, Winter 2023, Fall 2021*
Teaching Assistant

Computer Architecture, Seoul National University *Spring 2016, Fall 2015*
Teaching Assistant

MENTORSHIP

Mentored undergraduate, MS, and PhD students at UC Santa Cruz on computer architecture and CPU performance research projects.

SERVICE

IEEE CAL: Reviewer, IEEE Computer Architecture Letters *2025, 2026*

MICRO'25: Scarab Microarchitectural Simulator Workshop Tutorial

ASPLOS'25: Artifact Evaluation Committee Member for International Symposium on Architectural Support for Programming Languages and Operating Systems

REFERENCES

Prof. Heiner Litz (University of California, Santa Cruz)
E-Mail: hlitz@ucsc.edu